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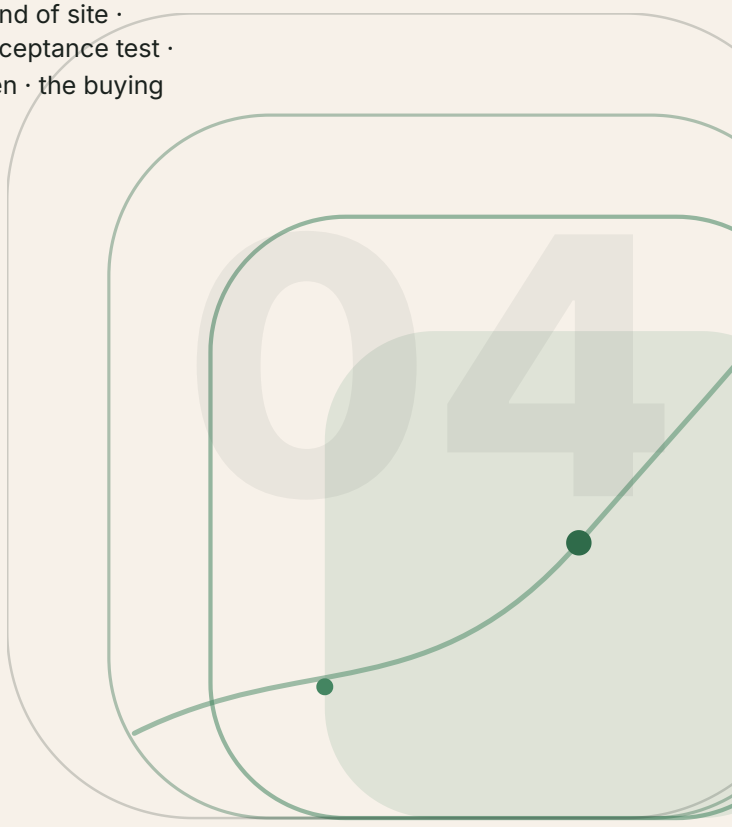
04

HARDWARE SHEET HW04 · QUALIFIED BY SPECIFICATION

The cellular router

The site's way to the internet where there is no cable the box may use: a 4G router with a data SIM.

QUALIFIES	Qualified by specification. Any 4G router for a DIN rail that takes a data SIM and has a network socket each for the box and the converter (or a small switch behind it).
USED IN	Kit F; optional in kits H, S and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who decides at the survey whether the site needs one
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW04

The cellular router

The box needs the internet, and it needs nothing else from it than to call out. Where the customer has a router the box may use, that is the best way and nothing is bought. Where there is none, as in a field or a pump station away from the buildings, a 4G router with a data SIM is the site's way to the internet, and the box and the converter plug into it.

At a glance

WHAT IT DOES	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.
HOW IT QUALIFIES	**Qualified by specification.** Any 4G router for a DIN rail that takes a data SIM and has a network socket each for the box and the converter (or a small switch behind it).
SPECIFICATION	4G, DIN rail, data SIM
FITTED BY	The project manager.
PLANNING ESTIMATE (EUR)	150 to 200. Not a quote.
USED IN	Kit F; optional in kits H, S and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It is a 4G router that fits a DIN rail and takes a data SIM.
- ☐ It has a network socket for the box and one for the converter, or a small switch is fitted behind it.
- ☐ The supply it needs is known before it shares the 24 V supply: read its label.
- ☐ The SIM is a data SIM on a plan of 5 GB a month or more. Greecon supplies the SIM, charged at cost.

CHECK

Check the signal where the router will sit, at the survey: look at your phone there. A router in a steel enclosure in a pump house hears less than a phone in your hand.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	OPTIONAL	0 or 1	Only where the building has no internet the box may use. Most hotels have a router: then the box and the converter plug into that.
S Solar plant with grid connection	OPTIONAL	0 or 1	Only without usable internet. The hardware guide's own list for this kit does not show one; the network rules are the same as for every kit.
F Farm or greenhouse with irrigation	IN THE KIT	0 or 1	The likelier case: a field or a pump house seldom has a cable to the internet. The box and the converter plug into the router's sockets.
W Water tank and pump station	OPTIONAL	0 or 1	Only without usable internet: a pump station that stands away from the buildings.

A site with no internet at all is not sold. The box keeps about three days of readings, which is for outages, not for a site that is never connected.

How much data it uses

Plan for about **1.5 GB a month**. That assumes a hotel-size site (four devices, seven readings) sending one request every ten seconds, about 5 KB each with its headers (about 1.3 GB), and a report every 30 seconds (about 0.15 GB). A site with more devices sends more. It is an estimate: read the router's counter after a week and write the real figure down.

Fitting

PROJECT MANAGER

- [ ] Put the SIM in, and mount the router on the rail in the enclosure.
- [ ] Plug the box and the converter into its sockets.
- [ ] Check the signal on the router's own lights or page, where it sits, with the enclosure closed.
- [ ] Label it with the site name. Keep the SIM's number and its plan in the hardware register, not on the router.

In the platform

The router is not in the platform. What shows it is working is the box: step 3 of the setup list, **Install the box**, says **Done**, and the box stays **Online**.

Acceptance test

- [ ] With the box and the converter plugged into the router, the box comes **Online** within a minute.
- [ ] The internet-cut test of commissioning is done by taking the SIM out for ten minutes: `greecon-edge status` shows a queue, and the queue is empty a few minutes after the SIM is back.
- [ ] After a week, the router's data counter is read and written down, and the plan is checked against it.

What goes wrong

What you see	Likely cause	What to do
The box is Offline, and the power is on	No signal where the router sits, the plan's data is used up, or the SIM is not a data SIM	Look at the router's signal lights. Read its counter against the plan. A spare SIM is in the spares kit
The box goes Offline and comes back, again and again	A weak signal	Move the router's aerial outside the enclosure, as its manual allows. Readings are not lost: the box sends them when it is back
The data runs out before the month does	A site with more devices than the estimate assumes	Read the counter, and take a larger plan

More is in the runbooks (Project Manager Field Manual, PDF 02): R1 (a box is offline).

Not yet proven

- The 1.5 GB a month is worked out, not measured. No Greecon site has yet run on a cellular router.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	